

DHEERAJ BHUKYA

AI/ML Engineer | Generative AI | Agentic AI | RAG

✉ dheerajbhukya1@gmail.com ☎ +919492141238 🔗 [linkedin.com/in/dheerajbhukya](https://www.linkedin.com/in/dheerajbhukya)

Professional Summary

Specialized in **Generative AI, Large Language Models (LLMs), LangChain, LangGraph, and MCP-based Multi-Agent Architectures**, with strong expertise in **Advanced RAG, Hybrid Retrieval, Vector Search, BM25, Embeddings, and Cross-Encoder Reranking**. Experienced in **production-grade AI deployment on AWS Bedrock** and building scalable **LLM applications and Agentic AI systems**. Proficient across the end-to-end **AI/ML lifecycle**, including **Data Ingestion, Embeddings, Information Retrieval, Model Development, Prompt Engineering, API Development, Evaluation, Monitoring, and Production Deployment**, with a proven track record of delivering **reliable, scalable AI solutions for real-world production environments**.

Core Technical Skills

Generative AI & Agentic AI — LLMs | Generative AI | LangChain | LangGraph | MCP | Multi-Agent Systems | Agentic AI | Prompt Engineering | Tool Calling | Function Calling

Cloud, Deployment & MLOps — AWS Bedrock | Amazon SageMaker | Amazon S3 | AWS Lambda | Docker | Kubernetes | MLflow | CI/CD | MLOps | Amazon CloudWatch | AWS ECR

Machine Learning — Recommendation Systems | Forecasting | Classification | Clustering | Anomaly Detection | Scikit-learn | TensorFlow | PyTorch | Regression | Feature Engineering | Model Evaluation

Backend & Data — Python | FastAPI | PostgreSQL | MongoDB | Redis | Pinecone | ChromaDB | FAISS

RAG & NLP — Advanced RAG | Semantic Chunking | Hybrid Search | BM25 | Vector Search | Cross-Encoder Reranking | Query Rewriting | Embeddings | Sentence Transformers | NLP

Professional Experience

INTENSO TECH SOLUTION PRIVATE LIMITED, AI/ML Engineer 04/2024 – Present

- Architected a **production Generative AI platform** on **AWS Bedrock (GPT, Claude)** for **enterprise content automation, conversational assistants, and knowledge retrieval**.
- Built **Advanced RAG pipelines** — **semantic chunking, hybrid retrieval, and reranking** — across **Pinecone and ChromaDB**, improving **retrieval precision by 28%**.
- Developed **LangGraph multi-agent workflows (planner, retrieval, reasoning, reviewer, tool-execution)** with **shared state, checkpointing, and human-approval gates** for sensitive actions.
- Integrated **MCP servers** to connect **LLM agents** to **enterprise APIs and databases** via **typed tool schemas, access controls, and auditable execution**.
- Built reusable **prompt and context-management frameworks** with **grounding and fallback validation**, cutting **manual review effort by roughly a third**.
- Deployed and monitored services with **Docker and Kubernetes**, adding **structured logging, tracing, and CI/CD automation**.

Unizen Technologies Pvt Ltd, Machine Learning Engineer 05/2022 – 03/2024

- Built **real-time recommendation systems (collaborative filtering, content-based, hybrid ranking)**, lifting **recommendation accuracy by 19%**.
- Developed **Kafka and Spark Streaming** pipelines for **clickstream processing** and **low-latency prediction delivery**.

- Built **churn-prediction, lead-scoring, and customer-segmentation models** that contributed to a **measurable drop in churn**.
- Automated **training, validation, and drift monitoring** with **MLflow** and **CI/CD**; exposed **real-time scoring** through **FastAPI**.

Projects

Multimodal Healthcare RAG & Clinical Document Intelligence, *Advanced RAG · Healthcare*

- Built an **Advanced RAG platform** for **clinical guidelines, prescriptions, discharge summaries, and lab reports** using **OCR** and **layout-aware parsing**.
- Implemented **Hybrid Retrieval** using **Vector Search + BM25, metadata filtering, and Cross-Encoder Reranking** to generate **source-grounded answers** with **page-level citations**.
- Designed a **LangGraph workflow** covering **document ingestion, retrieval, clinical summarization, safety review, and response validation**, using **MCP tools** for controlled access to **patient documents**.
- Implemented **AI safety and governance controls** including **RBAC, audit logs, confidence thresholds, and clinician escalation** for **high-risk queries**.

Tech Stack: Python | FastAPI | LangChain | LangGraph | MCP | AWS Bedrock | Amazon Textract | ChromaDB | OpenSearch | PostgreSQL

Enterprise Multi-Agent Operations Copilot, *Agentic AI · MCP Automation*

- Designed a **supervisor-based LangGraph Multi-Agent System** with **planner, retrieval, data-analysis, action-execution, and reviewer agents** to automate **multi-step operational workflows and approval processes**.
- Built **MCP servers** for controlled access to **PostgreSQL, REST APIs, and ticketing systems**, implementing **RBAC, allowlists, and comprehensive audit trails**.
- Grounded **agent decisions** through a **RAG layer** using **hybrid retrieval and reviewer validation**, reducing **manual workflow steps by ~45% in controlled testing** while maintaining **human oversight** for **high-impact actions**.

Tech Stack: LangChain | LangGraph | MCP | FastAPI | AWS Bedrock | PostgreSQL | Redis | ChromaDB | Docker | Kubernetes

Education

Lovely Professional University, Bachelor of Technology

2022 | Phagwara Punjab India

Achievements

Recognized as a **Star Performer** for contributions to **AI-driven cloud automation** initiatives that helped reduce **infrastructure costs by 20%**.

Improved **retrieval precision by 28%** by engineering **Advanced RAG pipelines** incorporating **semantic chunking, hybrid retrieval, vector search, and cross-encoder reranking** across **Pinecone and ChromaDB**.

Automated the **Machine Learning lifecycle** through **training, validation, model monitoring, and drift detection** using **MLflow and CI/CD**, while exposing **real-time model scoring APIs** through **FastAPI**.